

Fizika

9-sinf

Laboratoriya ishi №5

Mavzu: **Shishaning nur sindirish ko‘rsatkichini aniqlash.**

Dars rejasi

Laboratoriya ishini bajarishda texnika xavfsizligi qoidalari bilan tanishish

Laboratoriya ishini bajarishda kerakli bo‘lgan jihozlar bilan tanishish

Laboratoriya ishini tajribada ko‘rsatish

Ishning maqsadi

- shishaning nur sindirish koʻrsatkichini aniqlashni oʻrganish.

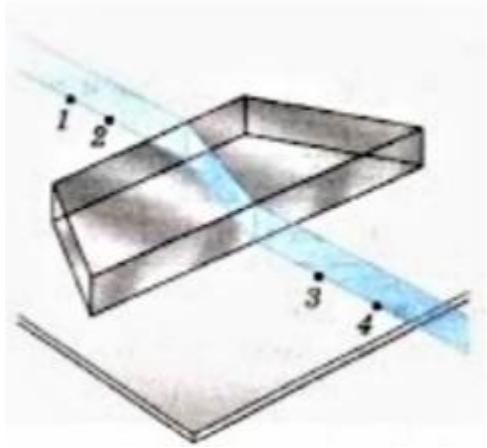
Kerakli jihozlar

- yorugʻlik manbayi, tirqishli toʻsiq, shisha prizma, ignalar, transportir, penoplast, oq qogʻoz, chizgʻich va qalam.

Ishni bajarish tartibi

1. Oq qog‘ozga shisha prizmani qo‘yib, tomonlari chizib olinadi.

2. Prizmani qog‘oz bilan birgalikda penoplast ustiga qo‘yiladi.



3. Yorug‘lik nuri hosil qilinadi. Yorug‘lik nuri prizmaga tushiriladi. Tushgan va singan nurlar yo‘liga ikkitadan ignalar qadaladi.

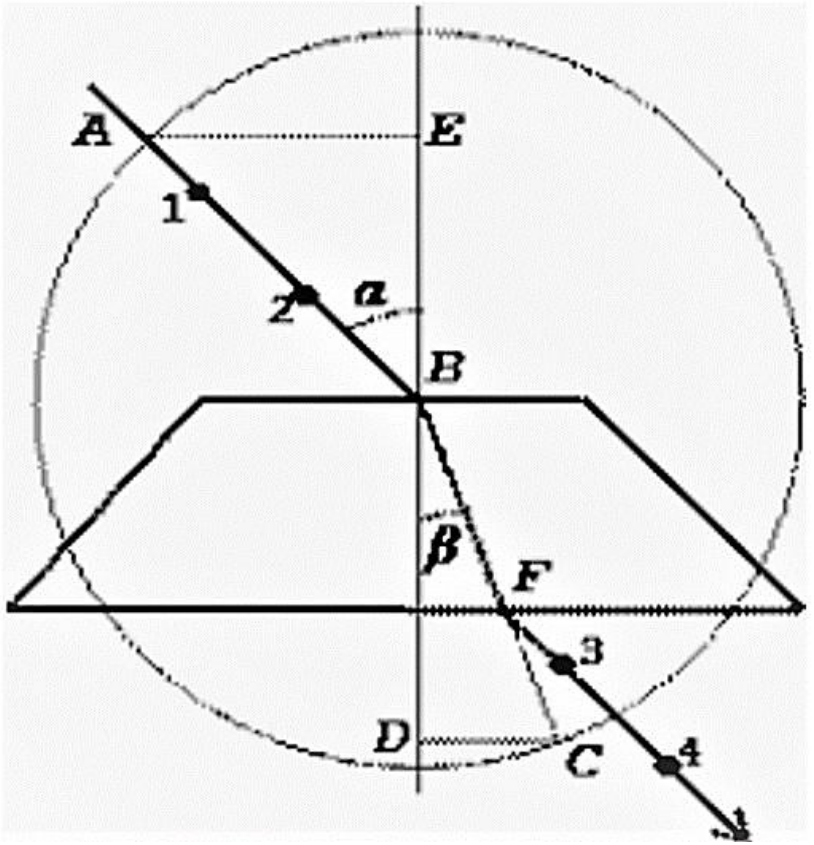
4. Shisha ostidan qog‘ozni olib unga quyidagi chiziqlar o‘tkaziladi.

5. Tushish va sinish burchaklari transportir yordamida o‘lchanadi.

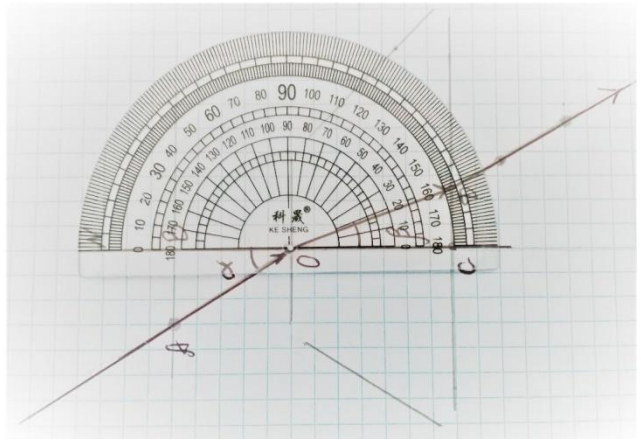
6. Quyidagi formula yordamida nur sindirish ko‘rsatkichi hisoblanadi.

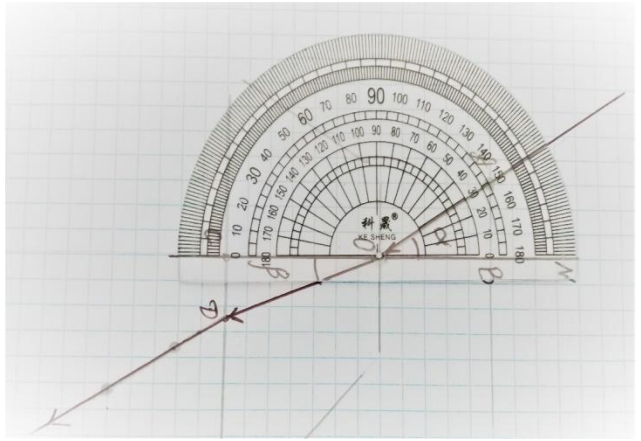
$$n = \frac{\sin \alpha}{\sin \beta}$$

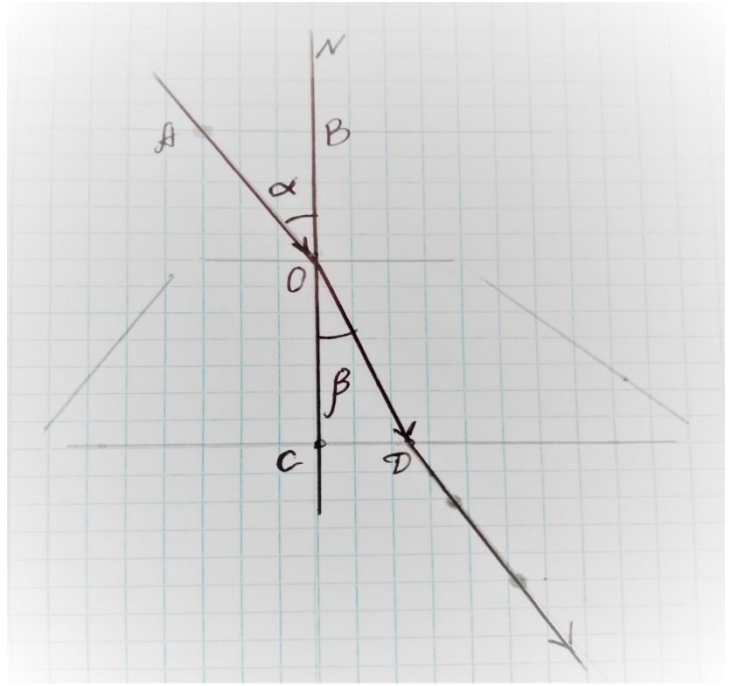
bunda α – shisha va havo chegarasiga nurning tushish burchagi, β – sinish burchagi.



5. O‘lchash va hisoblash natijalari: 1-usul.



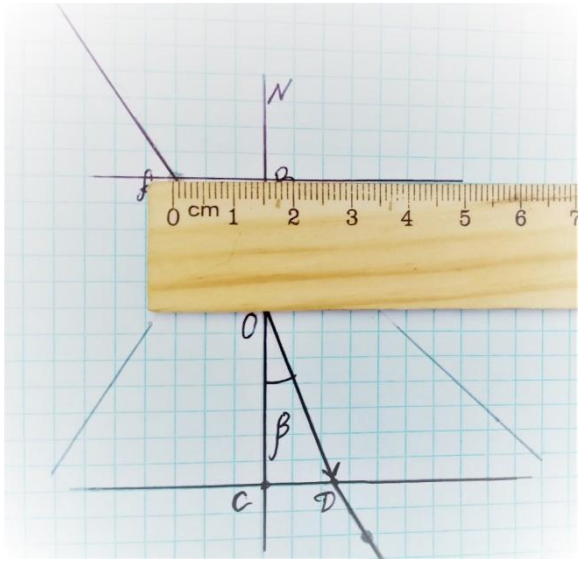


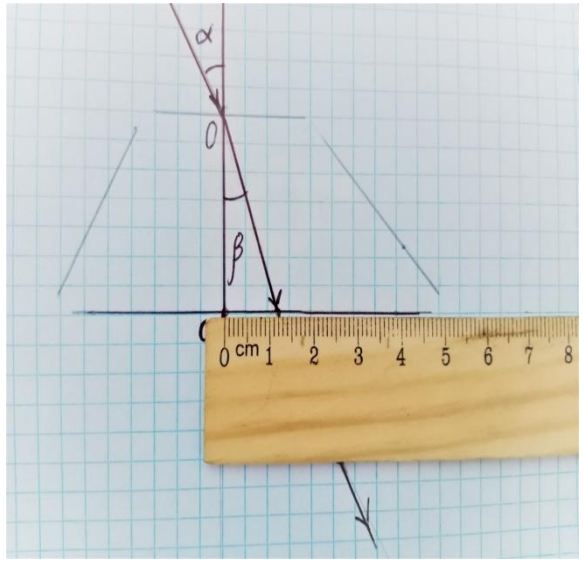


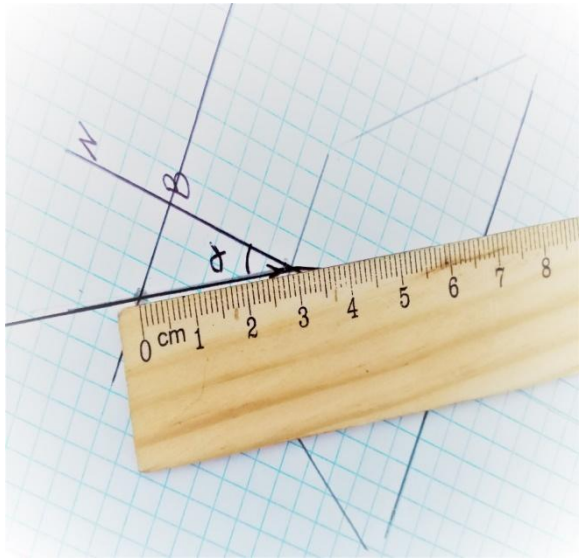
$$\alpha = 30^0, \quad \beta = 18^0.$$

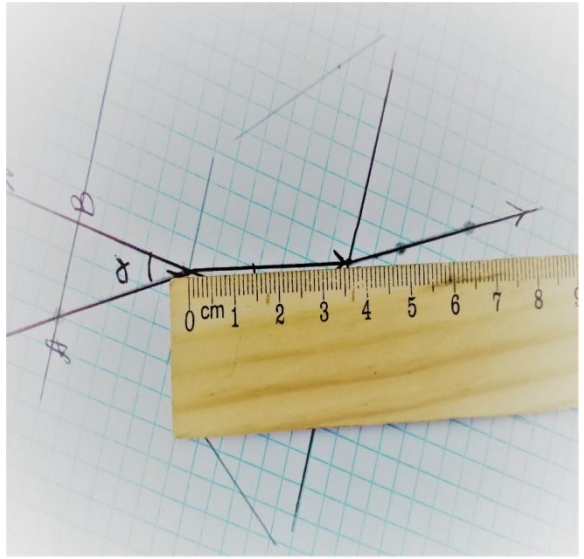
$$\sin 30^0 = 0{,}5\,; \qquad \sin 18^0 = 0{,}309$$

$$n = \frac{\sin 30^0}{\sin 18^0} = \frac{0{,}5}{0{,}309} = 1{,}6$$









2-usul:

$OA = 28\ mm$

$AB = 15\ mm$

$OD = 35\ mm$

$CD = 12\ mm$

$$\sin\alpha = \frac{AB}{OA} = \frac{15\ mm}{28\ mm} = 0{,}536$$

$$\sin\beta = \frac{CD}{OD} = \frac{12\ mm}{35\ mm} = 0{,}343$$

$$n = \frac{\sin\alpha}{\sin\beta} = \frac{0{,}536}{0{,}343} = 1{,}56$$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuga oid savollarga javob yozing. (103-bet)

2. Laboratoriya ishi natijalarini tahlil qiling va xulosa yozing.

3. Tajribadagi yorug‘lik nurini yo‘lini tahlil qiling, tushish va sinish burchaklarini ko‘rsating.